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International AS and A-level

Biology (9610)
BL01
Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL AS BIOLOGY (9610) BL01 - JANUARY 2025

Some students showed a good understanding of all areas of the unit, and applied biological terminology and concepts well. On the whole, straightforward recall-based marks were mostly scored. The more able students were able to score well on the applied questions. Exceptions were the structure of triglycerides and understanding of the terms 'intraspecific variation' and 'locus', where many students showed a lack of knowledge. Other weaker areas included questions that involved understanding standard deviation or graph interpretation. Most of the questions on the paper were attempted by the majority of the students.

QUESTION 01

Q1.1

Many students achieved both marks for this question. Some students did not score the second marking point (MP) because they were not specific enough and only called it a 'base' or incorrectly called it a nucleotide.

Q1.2

Most students scored three marks, although some tried to use the word 'extrons', which was not awarded. MP3 was the marking point least likely to be awarded, with incorrect answers 'alleles' or 'introns' being the two that appeared most often.

Q1.3

The vast majority of students scored two marks for the correct answer of 595. Some only achieved one mark due to incorrect rounding or incorrect number of decimal places, eg 594.69. After these types of mistakes, the next most common answer was 297, whereby students had incorrectly taken into account the effect of base pairing on length.

Q1.4

Few students achieved four marks, with many not mentioning the DNA unwinding and thus not achieving MP1. Another common mistake was not being clear that the DNA polymerase did not create complementary base pairs/join nucleotides to the template strand/join together free nucleotides.

Q1.5

Many students misinterpreted or did not read the question correctly so did not score marks, as they did not use their knowledge of enzyme structure. Some students did not think enzymes were used at all. A number of students did not mention the words 'active site' so could not be awarded MP1.

The idea of ‘fit’ or ‘complementary’ was needed for MP2. The idea of the enzyme being specific without it being qualified was not enough for the mark. ‘Binding’ was also not awarded, as this also does not contain the idea of complementarity.

QUESTION 02

Q2.1

The majority of students gained MP1 and correctly understood that the activation energy was lowered. However, few students also went on to score MP2, and most just referenced the formation of an enzyme-substrate complex instead. Some students referenced an alternative reaction pathway, but this was not deemed detailed enough to be awarded a mark.

2.2

Most students were able to identify the definition of quaternary structure.

2.3

The majority of students scored this mark. The most common mistake was to suggest that energy was made or produced, as this could not be awarded.

2.4

Many students lost marks for this question because they did not consider the command word and described rather than explained the results. MP1 was the MP most often scored, while few scored MP3. Some students attributed the plateauing to all the active sites on the catalase becoming saturated. Some students could not be awarded it because they did not talk about them levelling off at different times.

Q2.5

Many students understood the best way to keep the flask at 35 °C was to use a water bath. Students also needed the idea of the water bath being set at the optimal temperature to award MP1.

Most students were able to score MP2, but those who talked about the enzymes being denatured were not awarded, as this would only be likely at much higher temperatures. Answers that just included the idea of keeping the enzyme at the optimal temperature were also not awarded, as this concept was already introduced in the stem.

Q2.6

Some students lost marks for this question because their answers were not directional eg just referencing the concentration of hydrogen peroxide, rather than talking about increasing it. Answers that discussed increasing the temperature were not awarded marks.

Q2.7

Few students scored all three marks, as many did not gain MP1. The idea of repeating the experiment without the potato was a common idea, but this could only be awarded one mark as there was not enough detail.

Some students had the idea of heating the potato to denature the enzymes. Unless it was clear that the whole experiment was not being carried out at high temperature, this could also not be awarded.

QUESTION 03

Q3.1

Most students scored at least one mark, many of them both marks. The main mistake seen was being too vague, especially during MP1. Some students talked about hydrophobic tails and hydrophilic heads, but did not use the term phospholipids.

Q3.2

Apart from those mentioning the use of nitrogen or inert gases, the only answers that were accepted were those mentioning the addition of a layer of oil on top of the solution. This was because the idea of just sealing the conical flask lid would still have trapped a layer of oxygen next to the solution. Any mention of a vacuum was ignored, as this could have affected the cells.

Q3.3

The majority of students recorded a result within the range and thus scored two marks. However, some students lost a mark as their calculation showed incorrect numbers eg those without 1.9 as the denominator, even though the final answer was within the range of acceptable answers.

3.4

Some students lost marks by writing the ‘with oxygen’ answer in the ‘no oxygen’ space and *vice versa*.

Many students were able to correctly identify which was active and facilitated diffusion, but provided no evidence for their conclusions from the graph, so could only be awarded two of the four marks.

QUESTION 04

Q4.1

More than half of the students were not able to score all of the marks available for describing a triglyceride, with many getting the names of the bonds confused, or not remembering the names of the component parts. Some students confused triglycerides with other biological molecules.

Q4.2

Many students found this question difficult, especially counting the carbon atoms on the molecules. The most common mistake was with regards to lipid C. A number of students forgot to count the second carbon in the brackets, and thus counted 37 carbons, or forgot to double the number in the brackets and thus counted 21 carbons. Most students were still able to identify that lipid B contains the C=C and were awarded for it.

Q4.3

Many students correctly understood the connection between the presence of C=C bonds, unsaturation and decreased melting point. However, relatively few students were able to explain how the C=C bonds decreased the melting point, which is what the question required.

Q4.4

Less students scored this mark than expected, given that there were non-carbon to carbon double bonds present in the diagram. A mark was not awarded for mention of other possible double bonds present, without reference to oxygen or phosphorus as the other molecule involved.

4.5

Most students scored both marks for this question. The most common mistakes were forgetting to mention that water was needed too or getting the order of adding ethanol and water incorrect. In addition, some students did not achieve MP2 as they referred to a positive result as ‘cloudy’. This was not accepted if it was the only reference to the colour, as clouds are not necessarily white. A small number of students did not discuss the correct test, or included the idea of heating the emulsion test up, which is incorrect.

QUESTION 05

Q5.1

Many students achieved all three marks on this question, but the most common cause of lost marks was answers given to the incorrect number of decimal places. ‘Error carried forward’ marks were available for the surface area to volume ratio calculation if the student had miscalculated the volume, but only if it was to the correct number of decimal places.

Q5.2

The concentration of the HCl was the most often variable identified. Most students were able to correctly discuss why an increase in this would increase the rate of diffusion. The most common reason for loss of the second mark was because students did not give a directionality to their answer. For example, the idea that an increased temperature increases diffusion rate, rather than just stating that it affects it.

Some students discussed factors which had already been eliminated by the wording of the stem, eg anything related to diffusion distances/size/shape etc.

5.3

The most common mistake was to talk about variation in the volume etc, when that is the independent variable of the experiment, or mentioning the small sample size, when that was already excluded in the stem. Other things that were not awarded were mention of the concentration of HCl without having the idea that this is something that could be differing, or referring to the volume rather than the concentration of HCl.

5.4

References to water potential were ignored as this generally does not influence the rate of diffusion of HCl. Only answers that referred to the living cells as being smaller or having a larger SA:Vol were accepted. References to active transport were also not relevant as the question was about diffusion, and it was presumed that the concentration of HCl inside a cell is as low as that inside agar at the start of the experiment.

5.5

Most students were able to correctly identify that large animals have a low SA:Vol, and the impact this would have on diffusion. Answers that mentioned respiration were not awarded as this was covered in the stem.

5.6

The mark scheme for this question was very broad, so most students were able to achieve the mark. Those that didn't mostly gave answers that were too vague eg just writing 'lungs' or 'specialist gas exchange system'.

QUESTION 06

6.1

Less than half of the students scored the mark for this question. Those who gave ambiguous responses that could also be interpreted as interspecific were not awarded eg intrespecific. The most common incorrect answer was 'mutation'.

6.2

Most students were able to identify either a genetic or an environmental factor, although very few discussed the possibility of an interaction between the two.

6.3

This question was generally poorly answered, with many students not seeming to know the definitions of standard deviation or range. Some students who did pick up the marks for the standard deviation definition dropped the range definition mark. This is because they did not complete the comparison by including the reason why the range value would be larger i.e. because it is the difference between the maximum and minimum data points.

6.4

A fair number of students were able to identify whether standard deviations over-lapped. However, few students were able to coherently evaluate the conclusions that can be drawn from the overlap in standard deviation bars. MP2 was the most awarded mark. Mention of small sample sizes or only four species used had to be qualified with reference to representativeness or the equivalent, otherwise they could not be awarded.

A number of students mentioned the difference between the sample sizes of the different parrots without discussing any possible effect, if any, of this. A number of students confused the numbers above the error bars to be the size of the standard deviations, even though they were identified as sample size in the stem.

QUESTION 07

7.1

Most students scored this mark. Many students used the word ‘camouflage’ and were the mark. Those that had the idea of the insects looking like leaves, also needed the idea that this made them difficult to spot to be awarded.

7.2

Over half of the students achieved full marks. The main mistakes were including ‘family’ even though it was mentioned in the stem of the question. Many students also included ‘genus’. Some students tried to write the actual taxonomic groups that leaf insects belong to, eg Insecta, but as the word family was included in the stem, only the general names of the taxonomic groups were awarded.

7.3

Most students scored this mark. Many mentioned differences in physical characteristics. However, comments about size were ignored because this can be affected by the age of an individual, so is not a good tool for separating organisms into different species. Some students used a circular argument here i.e. they were originally classified into different species because their species names were different or even argued that it was because they were different sexes.

7.4

The majority of students achieved the marks, but some students didn’t score because they were not clear on what direction the result would be. Due to the phrasing of the question, answers along the lines of ‘to see if the offspring were fertile’ were accepted, but ‘to see if the offspring are not fertile’ was only accepted if it was qualified to explain that this would show that the species are different.

7.5

Many students didn’t score as they just repeated the question or directly quoted from the passage. Reference to sharing a recent common ancestor had to be worded in the direction of the information from the passage. General phrases such as ‘the more recent the ancestor, the more similar the DNA’ would not have been awarded the mark.

Q7.6

Most students who answered knew the correct response to this question. Although a surprising number of students did not attempt an answer. The most common incorrect answer was the word ‘taxonomy’.

Q7.7

A fair number of students lost marks through not reading the stem of the question carefully and including breeding experiments and/or DNA sequencing in their answers. Marks were only awarded for clear examples eg ‘compare the mRNA sequence’. Answers that involved morphology could not only reference highly variable traits such as size etc, as these might change throughout the lifetime of the species.

